

Generation Interconnection System Impact Study Report

For Queue Project Q13-0213

Purcell 138 kV

June 22, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0213. The Interconnection Customer proposes a 103.1 MW solar photovoltaic facility interconnecting at the Purcell 138 kV substation in the New Jersey area. The request was received on March 12, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 158 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,994,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Colorado to Yahara River 138 kV
Contingency: Multitrade Of Pittsylvania Lp to Sne - Sansum N Elec 138 kV line
Loading: 131.8% (Rating: 235 MVA)
DFAX: 7.7%
Pre-project loading: 128.4%
2. Facility: West Pt Municipal to Yahara River 138 kV
Contingency: Ft Sage to Wellsville 138 kV line
Loading: 126.2% (Rating: 244 MVA)
DFAX: 12.7%
Pre-project loading: 120.8%
3. Facility: Walnut-Engle Tap to Cabinet Gorge 138 kV Line
Contingency: Hawthorne to Purcell 138 kV
Loading: 102.7% (Rating: 254 MVA)
DFAX: 9.3%
Pre-project loading: 98.9%
4. Facility: Hawthorne to Russell Energy Center 138 kV
Contingency: Walnut-Engle Tap to Cabinet Gorge 138 kV
Loading: 101.3% (Rating: 174 MVA)
DFAX: 66.5%
Pre-project loading: 61.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Colorado to Yahara River 138 kV	Rebuild line with higher-capacity conductor	\$926,000	100% (\$926,000)
West Pt Municipal to Yahara River 138 kV	Rebuild line with higher-capacity conductor	\$1,438,000	100% (\$1,438,000)
Walnut-Engle Tap to Cabinet Gorge 138 kV Line	Reconductor line	\$1,109,000	100% (\$1,109,000)
Hawthorne to Russell Energy Center 138 kV	Reconductor line	\$521,000	100% (\$521,000)
	Total allocated to project		\$3,994,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.